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Research papers

Occurrence and controls of preferential flow in the upper stream of the Heihe River Basin, Northwest China

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ABSTRACT

Preferential flow (PF) has a faster migration speed and is an important process affecting hydrological, geochemical and ecological cycles. However, the factors that control PF during infiltration are not fully understood, especially in the data-scarce high and cold mountainous areas, where few studies have been focused on PF. We selected eight soil moisture observation stations in the upper reaches of the Heihe River Basin (HRB), Northwest China. During the growing season from 2014 to 2019, the soil moisture was monitored every 30 min at 5 depths (5, 15, 25, 40, and 60 cm) at each station in order to analyze the occurrence and spatial–temporal control of PF in a cold, mountainous environment. According to the difference in response time of shallower and deeper soil moisture to rainfall, infiltration events were classified into PF and matrix flow. Our results show that the frequency of PF events across all layers of these stations ranged from 0 to 22.28%. As depth increased, the average PF frequency gradually decreased, but the coefficient of variation of PF frequency increased. The effect of vegetation on the occurrence of PF was profound, because vegetation affects root distribution and fauna activities, which affects the occurrence of PF. Our results also show that the frequency of PF gradually decreases from scrubland, to grassland and meadow, and barren land. Low bulk density, low residual water content, high saturated hydraulic conductivity, and high soil organic carbon were found to be conducive to the occurrence of PF. In addition, we found that PF possessed a high degree of seasonal variability because of its strong dependence on soil moisture, rainfall and vegetation conditions. High rainfall was conducive to PF occurrence. In areas with high sand content, low initial soil moisture was conducive to PF occurrence (likely due to hydrophobicity), whereas high initial soil moisture was conducive to PF occurrence in areas with low sand content. Overall, vegetation, soil properties, and water conditions seem to jointly affect the factors that occur in spatial–temporal of PF, and the influence of initial soil moisture on PF varies across different sand content. Findings of this study reveal the occurrence and control factors of PF, which will improve our understanding of water cycling related to the infiltration process, recharge process and the runoff generation process at hillslope over the cold mountainous areas.

1. Introduction

Preferential flow (PF) refers to the phenomenon whereby a fluid bypasses most of the matrix when passing through a porous medium, and selects a preferred path in order to pass through the medium at a faster speed (Flury et al., 1994; Lin, 2010; Guo and Lin, 2018). A large number of studies have shown the ubiquity of PF in various landscapes (Jarvis, 2007; Koestel and Jorda, 2014; Wiekenkamp et al., 2016; Guo et al., 2018; Jarvis, 2020). Due to the faster migration speed of PF, it is considered to be an important factor affecting the hydrological cycle (Guo et al., 2014; 2019), soil erosion (Tao et al., 2020), hillslope stability (Kukemilks et al., 2018; Wang et al., 2019), farmland productivity (Zhang et al., 2015; Zhao et al., 2015b) and many other geochemical and ecological processes (Kan et al., 2020; Wang et al., 2020; Li et al., 2021).

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A soil moisture sensor network provides a method for long-term, large-scale, and continuous monitoring of PF in a natural state (Ochsner et al., 2013; Bogena et al., 2018; Guo et al., 2018, 2019; Zhang et al., 2018; Tang et al., 2020). This method has been widely used to study the spatial and temporal variability of PF at a watershed scale (Demand et al., 2019; Tang et al., 2020).

Many studies have found that the occurrence of PF is affected by various spatial and temporal factors (Wiekenkamp et al., 2016; Guo and Lin, 2018). Spatial factors influencing PF mainly include vegetation, topographic characteristics, and soil texture and the hydraulic properties, which together provide a flow path for the occurrence of PF (Wiekenkamp et al., 2016; Guo and Lin, 2018; Demand et al., 2019; Tang et al., 2020; Zhang et al., 2020). The most important temporal factors affecting PF are rainfall (water input) and initial soil moisture (Wiekenkamp et al., 2016; Guo and Lin, 2018; Demand et al., 2019; Gao et al., 2019). These factors jointly control the occurrence of PF, but the influence of these factors on PF is usually station-specific and may even be contradictory under different conditions (Guo and Lin, 2018).

The influence of control factors on PF varies from site to site (Guo and Lin, 2018). For soil porosity, intuitively, it should be evident that higher soil porosity will enhance PF, since macropore is the main path of PF (Mossadeghi-Björklund et al., 2016). However, a high soil porosity denotes that the surface area between the pores and the matrix is large, and a large amount of water will penetrate into the matrix, which is less conducive to the occurrence of PF (Jarvis, 2007; Larsbo et al., 2016). Many soil properties (bulk density (Bulk), saturated hydraulic conductivity (K_S), and soil organic carbon (SOC)) related to soil porosity also have similar contradictions in their influence on PF (Gjettermann et al., 1997; Jarvis, 2007; Larsbo et al., 2014, 2016; Gao et al., 2018; Guo and Lin, 2018). For example, Bulk, which is related to water transmission and water capacity, reflects not only the abundance of macropores but also the density of the matrix material (Nimmo, 2021). Hence, the relationship between Bulk and PF is diverse across different studies (Koestel et al., 2012, 2013; Mossadeghi-Björklund et al., 2016; Wiekenkamp et al., 2016). In addition, the influence of soil moisture on PF also varies across different research areas. Drier soils in areas with high sand content are conducive to the occurrence of PF, but PF in other areas is positively correlated with the initial soil moisture (Guo and Lin, 2018; Demand et al., 2019; Tang et al., 2020). Overall, these factors have different effects on PF at different study areas.

Current researches on PF mainly focus on farmland (Koestel et al., 2013; Larsbo et al., 2014, 2016) and humid mountains or hills (Wiekenkamp et al., 2016; Demand et al., 2019; Guo et al., 2019; Tang et al., 2020) few studies have investigated the occurrence and control of PF in cold, mountainous areas (Li et al., 2013; Hu et al., 2016). In an arid endorheic river basin, the water resources of the entire basin mainly originate in the headwater mountainous regions (Zhang et al., 2016; Gao et al., 2020a). PF is of great significance to the hydrological process in such study areas (Xiao et al., 2020). Jiang et al. (2021) found that PF is the main source of groundwater recharge (accounting for 66.4% of the total recharge) for grassland in the northern Qinghai-Tibet Plateau. Gao et al. (2020b) found that the deep soil moisture in the degraded alpine meadow was supplied by PF. Paquette et al. (2018) believed that the PF path is the principal hydrological pathways of headwater catchments in permafrost areas. In addition, PF is of great significance to the process of snowmelt infiltration into frozen soil (Stahli et al., 1996; Mohammed et al., 2021). However, the mechanism of PF control and occurrence in alpine regions are still poor.

Understanding the occurrence and control of PF is important for understanding the flow process of PF and can improve water resource management. Therefore, the main aim of this study is to identify and compare the occurrence of PF using a soil moisture network in a mountainous area. We analyze the effects of spatial factors (vegetation and soil characteristics) and temporal factors (initial soil moisture and rainfall) on the occurrence of PF. More specifically, we attempt to answer the following questions: Are there any PF control factors that are specific to cold, mountainous areas? How do these control factors affect PF occurrence in such areas, given that the influence of soil properties on PF can be contradictory? Clarifying the main mechanisms that control the initiation, persistence, and dynamics in spatial and temporal of PF in the alpine region.

2. Material and methods

2.1. Study area

This research was carried out in the upper reaches of the Heihe River Basin (HRB), China's second-largest inland river basin (or terminal lake) (Cheng et al., 2014). It is located in the Qilian Mountains (9729´–10132´E, 3743´–3939´E) on the northern margin of the Qinghai-Tibet Plateau, and has an area of over 27×10^3 km^2 (Fig. 1a). The study area has a temperate semi-arid to semi-humid continental monsoon climate, and lies at an altitude of 2000–5500 m above sea level. According to the observed meteorological data from 1960 to 2012, the average annual temperature ranges from −3 °C to 7 °C, and the annual precipitation ranges from 200 mm to 700 mm (Geng et al., 2014). Due to the high-altitude semi-arid climate, the temperature and precipitation have large temporal and spatial variability within the study area (Qi and Cai, 2007; Zhang et al., 2016). Most of the precipitation occurs from June to August, accounting for about 65% of the annual precipitation (Zhang et al., 2016). The strong vertical difference in temperature and precipitation has also led to a zonal vegetation distribution consisting of forests, alpine meadows, grasslands, scrublands, and sparse vegetation (Yang et al., 2015; Tian et al., 2019; Zhang et al., 2020). The main land cover types in the study area are grassland, forest, water body, and

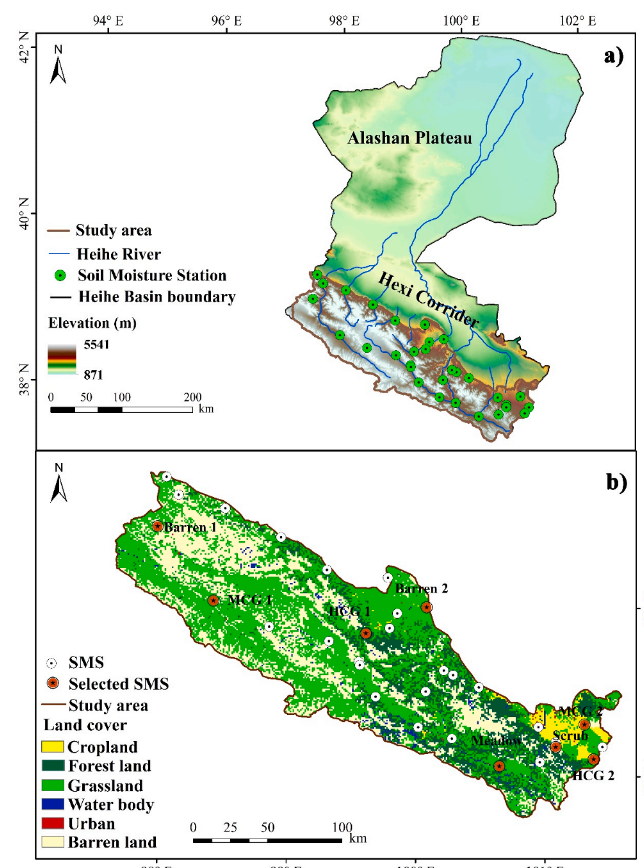

Fig. 1. (a) Location of the study area and distribution of the soil moisture stations (SMS). (b) Distribution of the selected SMS and land cover types in the study area.

barren land (Fig. 1b), and the main soil texture categories are silt loam, silt, and sandy loam (USDA classification) (Tian et al., 2017).

2.2. Soil moisture network

In order to investigate the dynamically changing characteristics of soil moisture in the upstream stream of the HRB under different environmental conditions, a long-term soil moisture observation network (including 32 soil moisture stations) was established in July 2013 (Fig. 1a). These stations include locations with different land cover (forest, grassland, meadow, and barren land) and soil types (silt loam, silt, and sandy loam), and different altitudes within the upper reaches of the Heihe River (Zhang et al., 2017; Tian et al., 2019). At each station, soil pits with a depth of 70 cm were excavated, and 5TE sensors (Decagon Devices Inc., Pullman, USA) were inserted horizontally (to avoid the impact on vertical water flow) at 5, 15, 25, 40, and 60 cm below the soil surface. Soil moisture was measured at a temporal resolution of 30 min, which is sufficient for studying soil hydrological processes in most cases (Lozano-Parra et al., 2016; Tian et al., 2019). At the depth where the sensors were installed, undisturbed and disturbed soil samples were collected with metal cylinders and zipped bags, respectively. We attempted to install the sensors at each observation station in vertical arrays, but this was not always possible due to the influence of rocks, roots, and channels. Therefore, there is a slight horizontal deviation relative to the ideal vertical array (Wiekenkamp et al., 2016). After installation, the excavated pit was refilled with the original soil and compacted to the initial state, layer by layer, in order to restore the natural soil matrix structure as much as possible. After sensor installation, we measured the station-related parameters (including land cover type, slope, aspect, and rooting depth). The collected soil samples were used to determine the key soil properties of each soil moisture observation station (including Bulk, soil porosity, K_S, SOC, soil–water retention curve, and soil texture). Details of the measurement processes of the above soil properties can be found in Tian et al. (2017).

2.3. Rainfall and NDVI

Under normal circumstances, high-precision rainfall data are obtained through the deployment of ground-based meteorological observation stations (Guo et al., 2018; Tang et al., 2020), but their spatial coverage is limited. Especially in mountainous areas, such stations are scarce, and it is more difficult to measure rainfall on a large scale (Yang et al., 2013; Zhao et al., 2015a). Therefore, in order to analyze the impact of rainfall on the occurrence of PF, we extracted rainfall data from the reanalysis data set (China Meteorological Forcing Dataset, CMFD) (Yang et al., 2010; He et al., 2020). CMFD is China's first near-surface meteorological raster dataset with a high temporal and spatial resolution (time resolution is 3 h, spatial resolution is $0.1^\circ \times 0.1^\circ$), and it is widely used in China (Meng et al., 2021; Zhang et al., 2021).

The monthly cumulative rainfall (P_m) of each station was calculated as following:

$$P_{mi} = \sum_{i,y=2014}^{i,y=2018} P_{y,i} \tag{1}$$

In the above equations: y is the year; i is the month (including the May, June, July, August, September and October); $P_{y,i}$ is the monthly cumulative rainfall.

In order to reflect the surface vegetation conditions of our observation stations, we extracted the Normalized Difference Vegetation Index (NDVI) during the growing season from 2015 to 2019 (except for 2016) from the surface vegetation index data of the Qilian Mountains, and used their maximum values as the NDVI of the corresponding station (Cihlar et al., 1994; Huete et al., 2002). The time resolution of the data set is 1 month, and the spatial resolution is 30 m × 30 m. Both the rainfall data and NDVI are from the National Tibetan Plateau Data Center (https://data.tpdc.ac.cn/en/). The monthly average NDVI ($NDVI_m$) of each station was calculated as following:

$$NDVI_{mi} = \frac{NDVI_{2015,i} + NDVI_{2017,i} + NDVI_{2018,i} + NDVI_{2019,i}}{4} \tag{2}$$

In the above equations: i is the month; $NDVI_{2015,i}$ is the NDVI value for a certain month in 2015.

2.4. Data analysis

2.4.1. Soil moisture

In a recent review, Nimmo (2021) concluded that the continuity of the flow path together with the properties and soil moisture of the surrounding matrix material are the main controlling factors of PF. To study the temporal-spatial control factors of PF in cold, mountainous areas, we selected a subset of stations with small slope and with different land covers (Demand et al., 2019; Tang et al., 2020). We selected stations with a relatively small K_S to ensure a long response time difference between the upper and lower soil layer moisture. We also chose stations where the data gap during 2014 to 2019 was<3 months. In the end, we selected 8 soil moisture stations, including those located on scrubland (Scrub), high coverage grassland (HCG1 and HCG2), medium coverage grassland (MCG1 and MCG2), meadow (Meadow), and barren land (Barren1 and Barren2). The locations and basic characteristics of these soil moisture stations are shown in Fig. 1 and Table 1.

We have collected soil moisture data since 2013, with a time resolution of 30 min. In addition, using the method provided by Decagon (Cobos and Chambers, 2010), soil-specific calibration was performed on each station (Tian et al., 2019). In order to fully restore the soil to its natural state and to eliminate the influence of freeze–thaw cycles on soil moisture, we selected only the growing seasons in each of the years from 2014 to 2019 (from May 1 to October 15 each year (Liu et al., 2015)) as the study period. Prior to subsequent data analysis, a detailed data quality control check was performed following the procedures of Wiekenkamp et al. (2016) and Tian et al. (2019). The specific steps were: 1) outliers were removed using quantitative plausibility checks (removing any value beyond the range of 0–90%, plus peaks and unreasonable fluctuations of soil moisture); 2) through visual data inspection, any unreliable data caused by equipment problems (such as insufficient battery power) was eliminated. In order to keep the infiltration events of each layer consistent, only periods when sensors in all five layers had recorded an observation were selected for subsequent analysis. In

Table 1
Basic characteristics of the selected soil moisture stations.

Soil Moisture Station	Stope (°)	Aspect (°)	Elevation (m)	Soil Texture	K_S (m/d)	NDVI
Scrub	19	78	3146	Silt Loam	2.15	0.64
HCG1	9	–	2556	Silt Loam	0.68	0.34
HCG2	6	65	2558	Silt Loam	0.31	0.58
MCG1	0	–	3317	Silt Loam	0.62	0.20
MCG2	0	–	2601	Silt Loam	0.67	0.41
Meadow	0	–	3105	Silt Loam	0.37	0.59
Barren1	17	197	3117	Silt Loam	0.21	0.17
Barren2	14	–	1827	Silt Loam	0.17	0.15

Note: HCG and MCG represent high-cover grassland and medium-cover grassland, respectively. Soil texture is analyzed according to the USDA soil classification scheme. K_S is the average saturated hydraulic conductivity of each station profile.

addition, the monthly average soil moisture in each station profile (SM_{mp}) was also calculated as following:

$$SM_{mpi} = \frac{\sum_{l=1}^{5} SM_{mli} \times L}{\sum_{l=1}^{5} L} \tag{3}$$

With

$$SM_{mli} = \frac{\sum_{y=2014,i}^{y=2019,i} SM_{yml}}{6} \tag{4}$$

In the above equations: y is the year; i is the month; SM_{yml} is the monthly average soil water content of the l-th layer in a certain month of a certain year; SM_{mli} is the monthly average soil moisture of each layer; L is the thickness of the soil layer.

Because the SM_{mp} of different stations are very various, in order to facilitate the comparison of the monthly changes of the SM_{mp} of different stations, calculated the ratio of SM_{mp} to the average soil moisture content (RSM_{mp}) of the profile:

$$RSM_{mp} = \frac{SM_{mp}}{SM_{ya}} \tag{5}$$

In the above equations: SM_{ya} is the average soil moisture of the profile during the growing season from 2014 to 2019.

2.4.2. Determination of the infiltration event

According to the definition of infiltration events by Tian et al. (2019), each infiltration event was determined by establishing its start and end times. For a soil moisture sequence, if its soil moisture increased continuously, the start time of the increase in soil moisture is defined as the starting time (Ts) of the infiltration event, and the end time of moisture increase is defined as the end time (Te) of the infiltration event. Any infiltration events occurring within six hours of one another (i.e., the time interval between the end of the previous event and the start of the next event was<6 h) were merged into the same event (Lozano-Parra et al., 2015; Lozano-Parra et al., 2016; Tian et al., 2019). In order to eliminate any increase in soil moisture caused by sensor errors, infiltration events in which the difference in soil moisture between Te and Ts of the event was<0.3% were removed (Rosenbaum et al., 2010; Lozano-Parra et al., 2015; Lozano-Parra et al., 2016). Finally, any soil moisture increases caused by fluctuations in soil temperature were eliminated using temperature-corrected soil moisture data, following the methodology of Saito et al. (2013) and Kapilaratne and Lu (2017). The identification of infiltration events was performed automatically using dedicated Matlab scripts.

We also calculated the soil water storage increment at each depth as follows (Tian et al., 2019):

$$SWS = L\sum_{Ts}^{Te} \Delta\theta_{i+}^{j} \tag{6}$$

With

$$\Delta\theta_{i+}^{j} = \begin{cases} \Delta\theta_i^j, & \Delta\theta_i^j > 0 \\ 0, & \Delta\theta_i^j \le 0 \end{cases} \tag{7}$$

In the above equations: SWS is the soil water storage increment (cm); L is the thickness of the soil layer (cm); $\Delta\theta_i^j = \Delta\theta_{i+\Delta t}^j - \Delta\theta_i^j$; $\Delta\theta_i^j$ is the change of soil moisture (vol. %).

2.4.3. Event classification

After determining the infiltration event of each layer of soil, the response time series of the specific location was used to classify the flow behavior. PF was defined as non-sequential response of the soil moisture, that is, for the same rainfall, the response time of deep soil was faster than that of shallow soil (Graham and Lin, 2011; Wiekenkamp et al., 2016). In addition, due to the high flow velocities of PF, it might also result in a sequential response with a very small difference in infiltration event response time (Hardie et al., 2013; Wiekenkamp et al., 2016; Demand et al., 2019). However, there is no uniform and effective method to determine the moving speed of the wet front corresponding to PF (Hardie et al., 2013; Demand et al., 2019). Moreover, as the soil moisture was monitored every 30 min, and it is impossible to obtain an accurate moving speed of the wet front (Jin et al., 2018). Hence, we only defined PF events as those with the same response time or non-sequential response for deeper and shallower soil infiltration events. According to the above principles, we can use the soil moisture data of two adjacent layers to determine whether PF occurs between the two layers. We can classify four types of response scenarios:

(a) layer 1st-2nd (the soil moisture response time of the second layer is earlier than or equal to the first layer);
(b) layer 2nd-3rd (the soil moisture response time of the third layer is earlier than or equal to the second layer);
(c) layer 3rd-4th (the soil moisture response time of the fourth layer is earlier than or equal to the third layer);
(d) layer 4th-5th (the soil moisture response time of the fifth layer is earlier than or equal to the fourth layer.

The classification of all infiltration events was automatically implemented in Matlab. An example of non-sequential response to an infiltration event is shown in Fig. 2.

In addition, the frequency of PF events (F_{Event}) and the proportion of SWS of PF events (P_{SWS}) can be calculated by taking the ratio of PF events to total infiltration events.

$$F_{Event} = \frac{N_{PF}}{N_{PF} + N_{SF}} \tag{8}$$

In the above equations: N_{PF} and N_{SF} are the number of PF events and sequential flow (SF) events, respectively.

$$P_{SWS} = \frac{SWS_{PF}}{SWS_{PF} + SWS_{SF}} \tag{9}$$

In the above equations: SWS_{PF} and SWS_{SF} are the total SWS of PF events and sequential flow (SF) events, respectively.

2.4.4. Statistical analysis

In this study, one-way analysis of variance (ANOVA) or Kruskal-Wallis test methods were used to test the differences in data (PF frequency, initial soil moisture, sand content and rainfall) (Lange et al.,

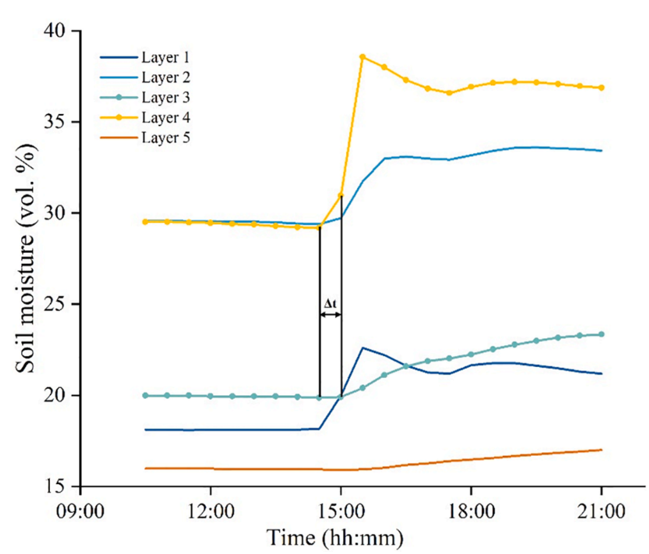

Fig. 2. Illustration of PF. The soil moisture is measured every 30 min. Layer 4 (30–50 cm) responds to soil moisture earlier than layer 3 (20–30 cm), indicating that water moves to the layer 4 earlier than the layer 3.

2008). Specific steps were: 1) a homogeneity test was used to determine whether the data satisfied the normal distribution and homogeneity of variance. 2) when the variance of the data set satisfied normality and homogeneity, ANOVA and a post-hoc Bonferroni test were used to test the difference between different sets of data. If these conditions were not satisfied, Kruskal-Wallis ANOVA with a post-hoc Dunn test were used for analysis. The statistics were implemented through SPSS (SPSS 25.0, SPSS Inc.) and Matlab (R2018a, The MathWorks).

3. Results

3.1. Frequency of PF

Table 2 shows the total number of infiltration events and the corresponding SWS of each layer of the eight selected soil moisture observation stations, as well as the number of PF events and their SWS. Combining data from all layers of the eight stations, we obtained a total of 2,288 infiltration events, with a total SWS of 5,404.86 cm, from 2014 to 2019. Among them, there were a total of 216 PF events with a SWS of 671.26 cm. The F_{Event} and P_{SWS} were 9.44% and 12.36%, respectively. On average, the SWS of a PF event was 3.11 cm, and the SWS of a SF event was 2.37 cm. This showed that the SWS of a PF event was greater than a SF event.

Table 2
Proportion of PF in different layers at the eight observation stations.

SMS	layer	Sum events	Sum SWS (cm)	PF events	SWS_{PF} (cm)	F_{Event} (%)	P_{SWS} (%)
Scrub	1-2	184	580.18	41	151.94	22.28	26.19
	2-3	152	666.24	28	117.87	18.42	17.69
	3-4	130	286.40	28	72.52	21.54	25.32
	4-5	116	206.02	12	29.88	10.34	14.50
	Total	582	1738.84	109	372.21	18.73	21.41
HCG1	1-2	58	133.38	9	18.13	15.52	13.59
	2-3	47	71.72	6	14.88	12.77	20.74
	3-4	33	46.57	4	3.47	12.12	7.45
	4-5	28	39.18	6	13.49	21.43	34.43
	Total	166	290.85	25	49.97	15.06	17.18
HCG2	1-2	103	289.48	6	13.47	5.83	4.65
	2-3	80	119.82	7	35.39	8.75	29.54
	3-4	55	83.02	4	13.28	7.27	16.00
	4-5	45	55.31	6	8.26	13.33	14.93
	Total	283	547.63	23	70.40	8.13	12.86
MCG1	1-2	96	372.56	13	61.36	13.54	16.47
	2-3	70	161.07	0	0	0	0
	3-4	44	76.62	1	1.56	2.27	2.04
	4-5	22	17.88	1	1.48	4.55	8.30
	Total	232	628.13	15	64.40	6.47	10.25
MCG2	1-2	149	522.56	9	43.36	6.04	8.30
	2-3	104	254.80	4	5.65	3.85	2.22
	3-4	44	102.53	1	10.84	2.27	10.57
	4-5	29	30.05	1	0.59	3.45	1.95
	Total	326	909.94	15	60.44	4.60	6.64
Meadow	1-2	85	167.18	4	7.50	4.71	4.48
	2-3	74	83.34	9	14.63	12.16	17.56
	3-4	35	25.67	0	0	0	0
	4-5	21	21.23	1	0.95	4.76	4.47
	Total	215	297.42	14	23.08	6.51	7.76
Barren1	1-2	93	296.25	6	13.34	6.45	4.50
	2-3	62	119.11	2	6.52	3.23	5.47
	3-4	34	50.27	1	0.42	2.94	0.83
	4-5	25	28.07	0	0	0	0
	Total	214	493.70	9	20.28	4.21	4.11
Barren2	1-2	93	399.50	2	8.35	2.15	2.09
	2-3	71	65.25	0	0	0	0
	3-4	59	40.01	3	1.57	5.08	3.92
	4-5	47	19.58	1	0.55	2.13	2.79
	Total	270	524.34	6	10.47	2.22	2.00
Total		2288	5430.86	216	671.26	9.44	12.36

Note: SWS_{PF} is the soil water storage increment of the PF event.

3.2. Discrepancies in PF between layers

The proportion of the various types of PF event scenarios outlined above is shown in Table 2. The relative frequency of PF in the second layer (layer 1–2) is 41.6%, in the third layer (layer 2–3) is 25.93%, in the fourth layer (layer 3–4) is 19.44%, and in the fifth layer (layer 4–5) is 12.96%. However, the frequency of PF in each layer does not decrease with depth for all stations. The frequency of PF at each depth is shown in Fig. 3. In these soil moisture stations, the variability of PF frequency at each depth was high (Table 3). The standard deviation was highest in layer 4–5 (F_{Event}: 7.11%, P_{SWS}: 11.28%), followed by layer 3–4 (F_{Event}: 7.08%, P_{SWS}: 8.77%), then layer 1–2 (F_{Event}: 6.84%, P_{SWS}: 8.21%), and finally layer 2–3 (F_{Event}: 6.70%, P_{SWS}: 11.17%). The variation of standard deviation with depth may be attributed to variations in root systems and soil properties with depth. The root penetration depth varies greatly between soil moisture stations (scrub: greater than 70 cm, meadow: 10 cm). Through observation of station profile photos (Supplementary Material, Fig. S1), we found that some stations contain gravel in their deep layer, but the distribution of gravel varies among stations. Macropores produced by roots (Johnson and Lehmann, 2006; Li et al., 2009; Jia et al., 2017; Wang et al., 2020) and gravel (Zhao et al., 2020) are important paths for PF. Hence, the greater standard deviation of PF in the deeper layers may be attributed to differences in the root system and gravel content of these layers across different stations.

3.3. General relevance of PF

PF was observed at each station, but the relative frequency of PF among these stations (the standard deviation of the PF frequency of these station profiles is 5.8%) and between different layers (Table 3) was very variable. The variability of PF between different landscape types and even within some stations was high, which highlights the heterogeneity of soil over large scales and shows the influence of small-scale soil properties on soil infiltration. However, despite the considerable variability, we still find that PF depends on particular spatial and temporal factors, which are analyzed below.

3.4. Spatial controls of PF

3.4.1. Land cover

At these stations, the F_{Event} in the profile and the P_{SWS} of PF are shown in Fig. 4. The frequency of PF gradually decreases from scrubland, to grassland and meadow, and barren land. In addition to bare land, P_{SWS} in other stations is greater than F_{Event}, indicating that the SWS of PF event is greater than SF event. In order to quantitatively investigate the influence of vegetation on the occurrence of PF, we analyzed the relationship between NDVI and the F_{Event}, and found an exponential trend between them (Fig. 5). The fitting results showed that the frequency of PF is positively correlated with NDVI ($R^2 = 0.242$, $p < 0.01$).

3.4.2. Soil properties

In order to explore the influence of soil properties on the occurrence of PF, we constructed a scatterplot of soil properties versus the frequency of PF (Fig. 6). Results showed that there were four soil properties that were significantly related to PF frequency, which were Bulk ($r = -0.687$, $p < 0.001$), K_S ($r = 0.603$, $p < 0.001$), SOC ($r = 0.676$, $p < 0.001$), and residual water content (θr: $r = -0.448$, $p < 0.01$). Our results showed that lower Bulk increased the incidence of PF, which is due to lower Bulk reflecting a larger number of macropores (Anderson et al., 1990). Moreover, macropores have a large influence on K_S (Bouma, 1982), which leads to a significantly positive relationship between K_S and PF ($r = 0.603$, $p < 0.001$). SOC is closely related to soil structure (pore volume, size distribution, and connectivity) and hydrophobicity (Guo and Lin, 2018), both of which have important effects on the occurrence of PF (Larsbo et al., 2016; Guo and Lin, 2018; Demand et al., 2019). Zhou et al. (2013) and Jiang et al. (2017) found that the greater the dry density of

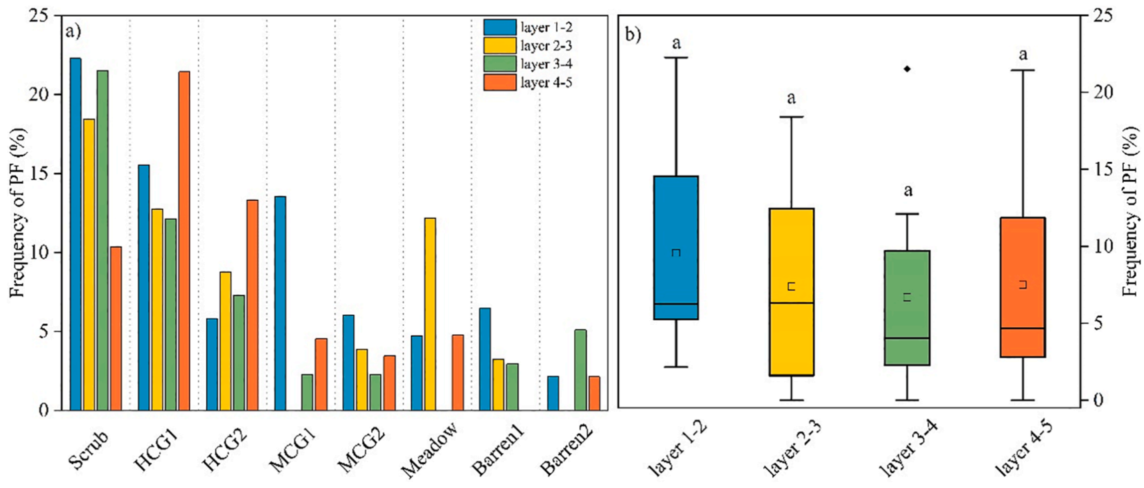

Fig. 3. Bar graph of PF frequency (a), and the boxplot of PF frequency of different layers (b).

Table 3
The PF frequency discrepancies of each layer of 8 soil moisture stations.

	Layer 1-2		Layer 2-3		Layer 3-4		Layer 4-5	
	F_{Event} (%)	P_{SWS} (%)	F_{Event} (%)	P_{SWS} (%)	F_{Event} (%)	P_{SWS} (%)	F_{Event} (%)	P_{SWS} (%)
Std. Deviation	6.84	8.21	6.7	11.17	7.08	8.77	7.11	11.28
CV	0.72	0.82	0.91	0.96	1.06	1.06	0.95	1.11

Note: F_{Event} is the frequency of PF events; and P_{SWS} is the proportion of SWS of PF events.

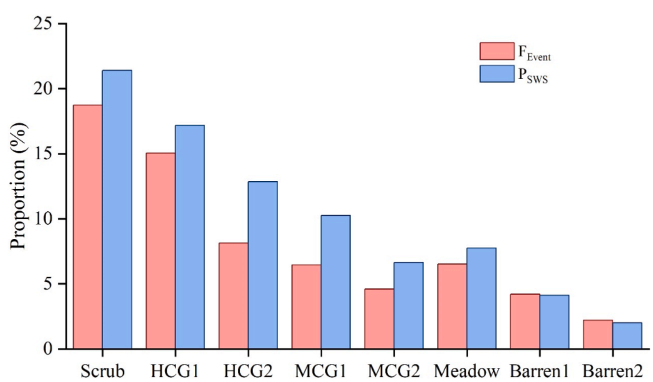

Fig. 4. The F_{Event} and P_{SWS} for scrubland (Scrub), meadow (Meadow), high coverage grassland (HCG1 and HCG2), medium coverage grassland (MCG1 and MCG2), and barren land (Barren1 and Barren2).

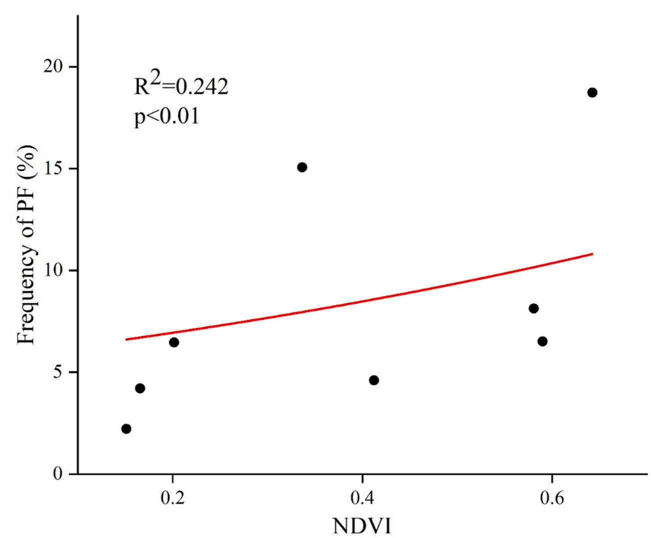

Fig. 5. Exponential fitting of PF frequency and NDVI. NDVI is the average value of each station during the 2015–2019 growing season (except for 2016).

the soil, the greater the θr. In Fig. 6, we also observed a positive correlation between θr and Bulk ($r = 0.756$, $p < 0.01$). In other words, the greater the θr, the greater the Bulk, and less PF will occur.

3.5. Temporal controls of PF

We found that the occurrence of PF is not only controlled by spatial factors, but also by time-variable factors. We analyzed the variation of PF frequency in different seasons (Fig. 7a). Our results show that the average relative frequency (the ratio between the sum of PF events in a certain month and all PF events.) of PF in summer (Jun: 20%; Jul: 24%; Aug: 17%) is much greater than in other months (May: 14%; Sep: 9%; Oct: 5%). The seasonal variation of PF frequency may be caused by seasonal differences in rainfall, vegetation, and soil moisture. Our study area has a temperate semi-arid to semi-humid continental monsoon climate, and more than 65% of the rainfall occurs in summer (Li et al., 2009). Fig. 7c also shows that the rainfall in summer is much higher than the other seasons. Seasonal variation in rainfall leads to seasonal differences in vegetation (Fig. 7b) and soil moisture (Fig. 7d). Similarly, in summer, the soil is wetter (RSM_{mp} is higher) and the vegetation is more florescent (the $NDVI_m$ is greater). The control of vegetation on PF has been described in section 3.4.1, so we do not repeat it here. However, the influence of initial soil moisture on PF varies between different landscape types. The influences of initial soil moisture and rainfall on the occurrence of PF are discussed below in more detail.

3.5.1. Initial soil moisture

We found that in some layers (Scrub: layer 4–5; HCG1: layer 3–4 and layer 4–5; HCG2: layer 3–4; Meadow: layer 2–3; Barren1: layer 2–3), the mean value of the initial soil moisture of PF events was lower than that of SF events (Fig. 8); the initial soil moisture of PF events was higher

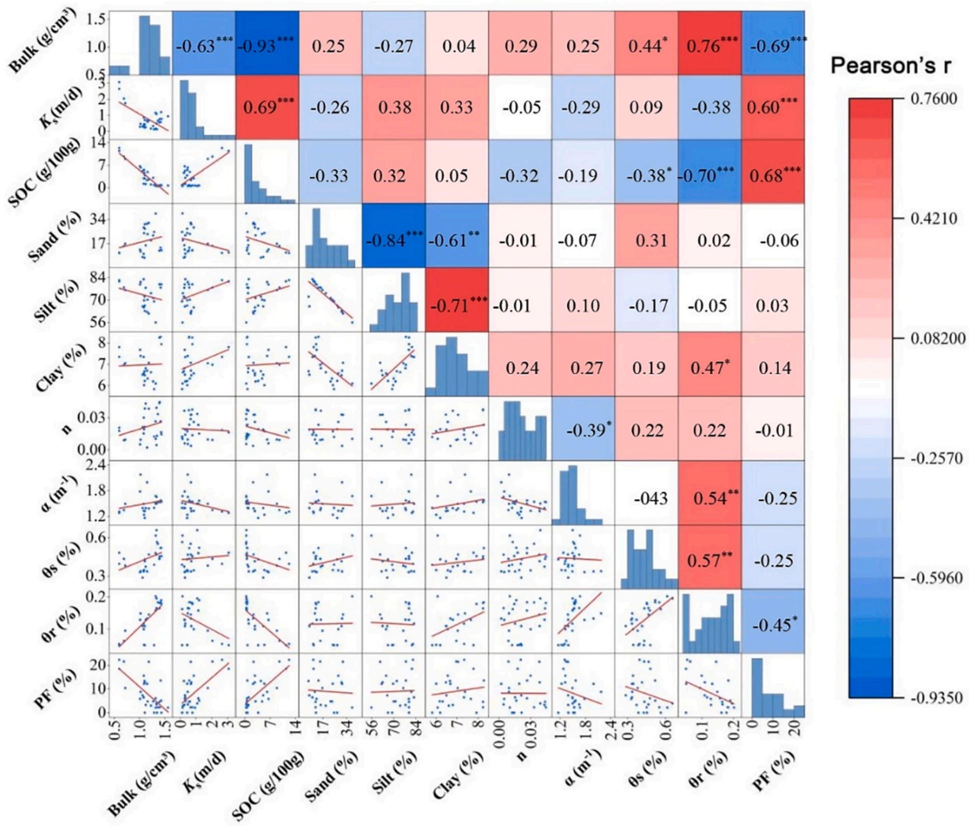

Fig. 6. Relationships between soil properties and the frequency of PF. n, α, θs, and θr are the parameters of the soil moisture retention curve. The parameters α and n are related to the inverse of air entry suction and pore-size distribution, respectively. θs is the saturated water content. *, ** and *** represent relationships that are statistically significant at the $p < 0.05$, $p < 0.01$, and $p < 0.001$ level, respectively.

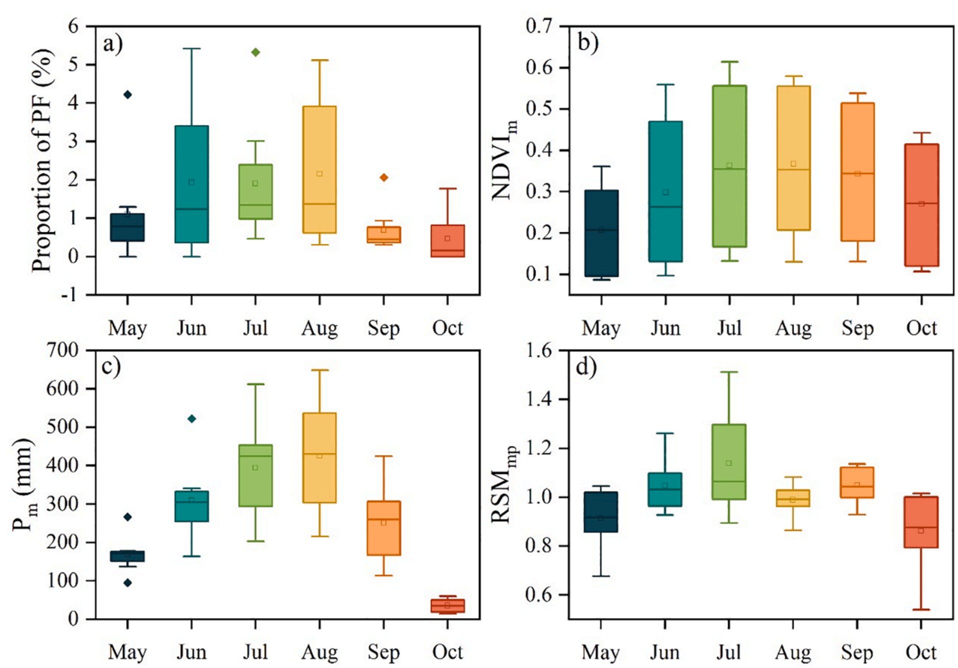

Fig. 7. Variations of PF frequency (a), NDVI (b), rainfall (c), and soil moisture (d) in different months.

than that of SF events in the other layers. We compared the sand content of the two categories (category 1: PF events had higher initial soil moisture than SF events, $\theta_{int,\ PF} > \theta_{int,\ SF}$; category 2: SF events had higher initial soil moisture than PF events, $\theta_{int,\ PF} < \theta_{int,\ SF}$) and found that the sand content of layers with $\theta_{int,\ PF} < \theta_{int,\ SF}$ was much higher than the sand content of the layers with $\theta_{int,\ PF} > \theta_{int,\ SF}$ (Fig. 9).

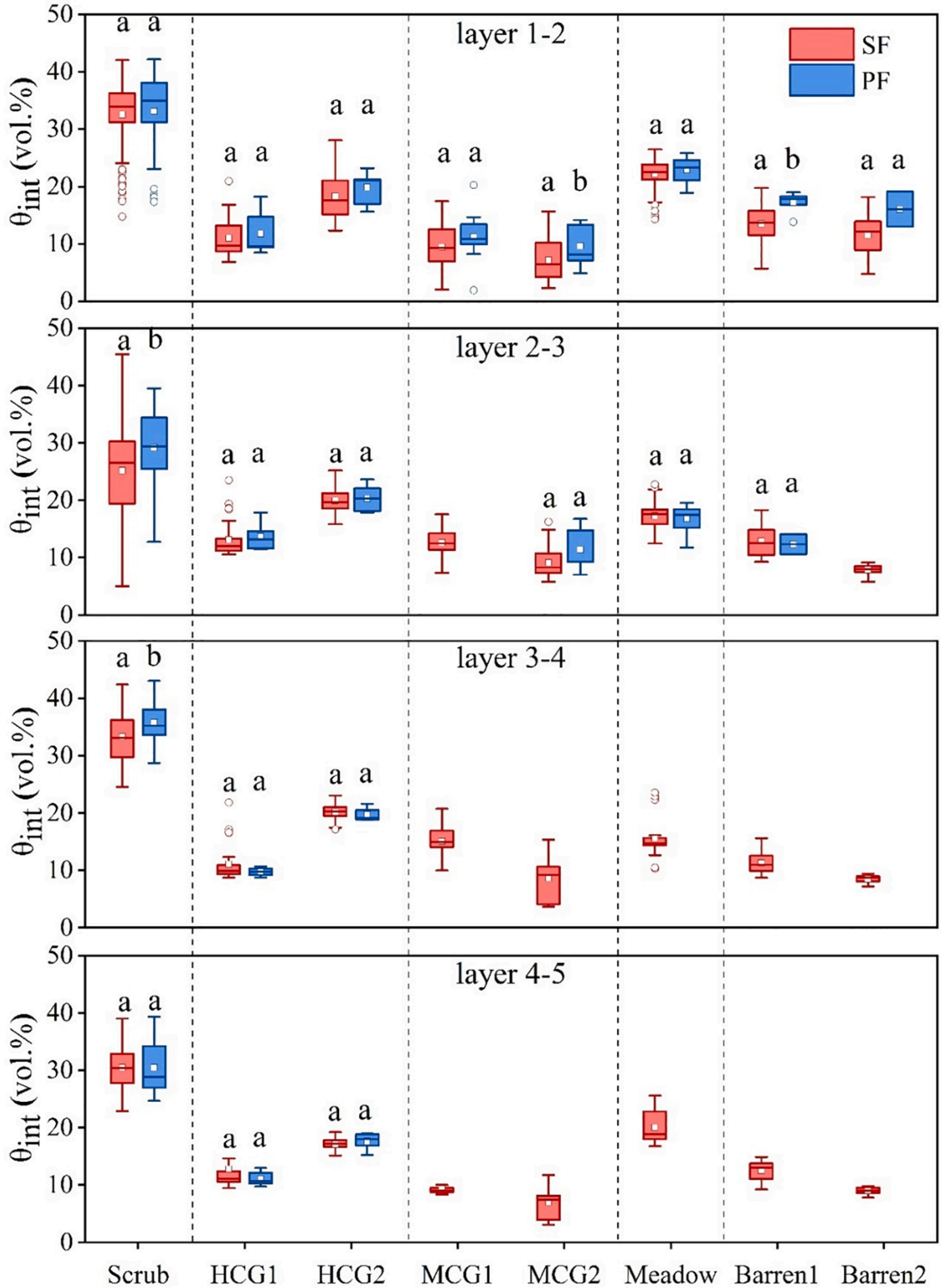

Fig. 8. Boxplot of initial soil moisture content of SF and PF events in each layer for the 8 soil moisture observation stations.

3.5.2. *Rainfall*

Many studies have recognized that the pattern of rainfall is the most important controlling factor of PF (Graham and Lin, 2011; Wiekenkamp et al., 2016; Guo and Lin, 2018). Commonly, short-duration and high-intensity rainfall are conducive to the occurrence of PF, while long-duration and low-intensity rainfall tends to uniformly moisten soil and produce SF (Guo and Lin, 2018). Given the absence of in-situ observations of rainfall in the soil moisture observation network, CMFD rainfall data was used to investigate the influence of rainfall on the PF occurrence. We calculated the amount of cumulative rainfall (the total rainfall of an infiltration event) corresponding to all infiltration events at each observation station (Fig. 10). Our results showed that the amount of cumulative rainfall of a PF event was significantly greater than that of a SF event ($p < 0.01$), which is consistent with previous data (Larsbo, 2011; Wiekenkamp et al., 2016).

4. Discussion

4.1. *Variability of PF*

Due to the strong heterogeneity of the factors that affect PF (Guo and Lin, 2018; Demand et al., 2019), PF has high variability among layers and among different observation stations. Based on the average value of PF of each layer, the frequency of PF tends to decrease with an increase in depth (Fig. 4 and Table 3), but this rule is not followed at all stations. Moreover, the deep soil properties (e.g., gravel content and root distribution) of these stations are quite different, hence the frequency of deep PF is more variable. Many studies have found that abundant roots and gravel are conducive to the occurrence of PF (Johnson and Lehmann, 2006; Li et al., 2009; Bogner et al., 2010; Jia et al., 2017; Luo et al., 2019; Wang et al., 2020; Zhao et al., 2020). We did not accurately measure the distribution of roots and gravel in the profiles of these

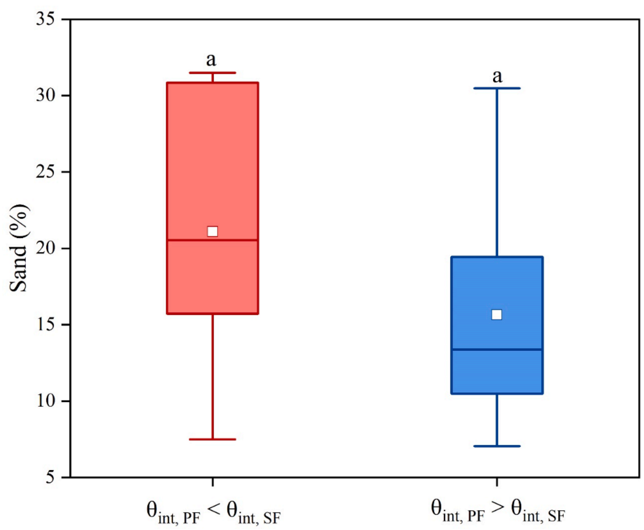

Fig. 9. Sand content of sites where the average value of the initial soil water content of the PF is low ($\theta_{int, PF} < \theta_{int, SF}$), and of sites where the average value of the initial soil moisture content of the PF is high ($\theta_{int, PF} > \theta_{int, SF}$).

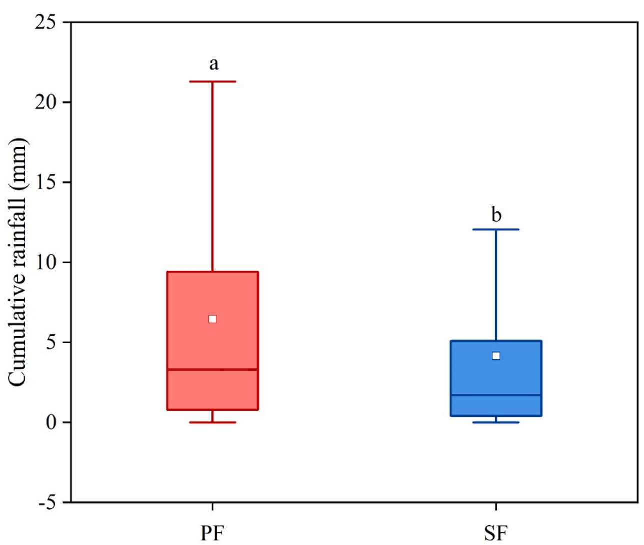

Fig. 10. Cumulative rainfall (the total rainfall of an infiltration event) for PF and SF events. Because the CMFD is only updated to 2018, the rainfall data in all graphs were from 2014 to 2018.

stations, hence we cannot quantitatively describe their impact on the occurrence of PF (Gjettermann et al., 1997; Guo and Lin, 2018).

Until now, most studies on PF have focused on locations with similar physical conditions (e.g., forests with temperate climates) (Demand et al., 2019). Our research includes landscape types (scrubland, grassland, meadow and barren land) other than forests, in cold, mountainous areas, and we find that the PF of these observation stations decreases as the vegetation coverage decreases. In our study, the occurrence of PF for a single soil moisture profile (2.22–18.73%) is different from other studies, and the maximum frequency of PF is much lower than in other studies (Liu and Lin, 2015; Demand et al., 2019; Tang et al., 2020). Compared with a forest, the root systems of the observation stations we chose are sparse, which is less conducive to the occurrence of PF. In addition, vegetation cover affects SOC, which in turn changes soil hydrophobicity and soil fauna activity, affecting the occurrence of PF (Wyatt et al., 2021). Moreover, the rainfall at our selected stations is also lower (Gjettermann et al., 1997; Guo and Lin, 2018). Rainfall is one of the most important factors affecting PF, hence PF occurrence at these observation stations is lower (Wiekenkamp et al., 2016).

In addition, the PF frequency of our stations are relatively small, which may also be caused by the method we defined PF events. The quantification of PF in current research is various. Liu and Lin (2015) defined PF as an event in which at least one deeper probe responds to rainfall input earlier than a shallower soil moisture, or an event in which the shallow and deep soil moisture respond sequentially but a soil moisture between them does not respond. Some studies also defined the event that the wetting front velocity is greater than a certain fixed value (experiment measurement (Hardie et al., 2013; Wiekenkamp et al., 2016) or model simulation (Demand et al., 2019)) as PF. Tang et al. (2020) defined any event in which soil moisture does not respond sequentially from the surface to the depth as PF. In this study, only events in which the deeper soil moisture response is earlier than the shallower layer or the shallower and deeper layers respond at the same time were defined as PF. The frequency of PF should be considered as a lower bound (i.e., probably underestimated the actual PF occurrence).

4.2. The influence of soil properties on PF

We found that low Bulk, low θr, high K_S, and high SOC were conducive to the occurrence of PF in this study. However, these results are different from some previous studies that have reached completely different conclusions (Koestel et al., 2013; Katuwal et al., 2015; Larsbo et al., 2016; Paradelo et al., 2016). Some previous studies found that low Bulk, high K_S and high SOC are not conducive to the occurrence of PF (Larsbo et al., 2014, 2016; Mossadeghi-Björklund et al., 2016). Moreover, there is a contradiction between the abundance of soil pores and the occurrence of PF. On the one hand, large pores are an important path for PF, and abundant pores are conducive to the occurrence of PF (Mossadeghi-Björklund et al., 2016). On the other hand, abundant pores mean that the contact area between the soil matrix and the outside world is larger, hence a larger amount of water can penetrate into the matrix, and this is not conducive to the occurrence of PF (Jarvis, 2007; Larsbo et al., 2016). Bulk, K_S and SOC are related to soil pores, so their impact on PF can likewise appear contradictory.

In addition, a significant relationship between soil texture and PF was not found in this study, although some studies have found that soil texture is an additional important factor affecting PF (Guo and Lin, 2018; Tang et al., 2020). This difference may be caused by our principles of soil moisture stations selection. We chose the main soil texture type (silt loam) of the soil moisture stations which we set up. In the same texture of soil, the difference between sand, silt and clay at different stations is very small, so the relationships between sand, silt, clay and PF frequency at these stations are not significant.

4.2.1. Bulk

Bulk has an important influence on water holding capacity and water transmission (Nimmo, 2021). We found that lower Bulk increased the incidence of PF, which can be attributed to the fact that lower Bulk reflects a larger number of macropores. Koestel et al. (2012) studied a large number of breakthrough curves and found that there was a negative correlation between Bulk and the strength of preferential transport (the average Bulk was 1.41 g/cm^3). Mossadeghi-Björklund et al. (2016) found that less PF occurs in the soil after compaction (the soil Bulk became larger). However, Koestel et al. (2013), Katuwal et al. (2015), and Paradelo et al. (2016) found that the strength of preferential transport was directly proportional to Bulk in farmland (the average Bulk was 1.47 g/cm^3, 1.55 g/cm^3 and 1.39 g/cm^3, respectively). Bulk has contradictory effects on occurrence of PF because Bulk reflects both the abundance of macropores, suggesting more PF when lower, and the density of the matrix material, suggesting more PF when higher (Nimmo, 2021). We compared Bulk with other studies and found that the PF frequency is proportional to Bulk at stations with higher Bulk (Koestel et al., 2013; Katuwal et al., 2015; Paradelo et al., 2016), while it is inversely proportional at stations with lower Bulk (Koestel et al., 2012; Mossadeghi-Björklund et al., 2016). We infer that this contradiction can be attributed to the existence of a threshold for the impact of Bulk on PF occurrence. When Bulk is below this threshold, the PF frequency is inversely proportional to Bulk, and when Bulk is above this threshold, the PF frequency is proportional to Bulk. Compaction (increased Bulk) will reduce soil porosity, which is unconducive to the occurrence of PF. However further compaction will strengthen the continuity of macropores, which is conducive to the occurrence of PF (Jarvis, 2007; Mossadeghi-Björklund et al., 2016).

4.2.2. SOC

SOC is closely related to soil structure (pore volume, size distribution, and connectivity) (Guo and Lin, 2018; Kerr and Ochsner, 2020; Jia et al., 2020). Our study indicated that PF frequency was proportional to the SOC (Fig. 7). Bundt et al. (2001) and Hagedorn and Bundt (2002) found that the SOC concentration in the PF path is 15–75% and 37–46% greater than the concentration in the matrix, respectively. Greater SOC enhances fauna activity (Gjettermann et al., 1997) and creates a soil environment conducive to the occurrence of PF (Arocena et al., 2010; Guo and Lin, 2018). Rahbeh (2019) found that soil organic matter is positively correlated with PF, and low soil organic matter is unconducive to the aggregation and formation of macropores. However, with a larger SOC, the flow capacity of micron-sized pores in the soil column is increased. This prevents flow in the larger pores from being activated, thereby significantly reducing the strength of PF (Ghafoor et al., 2013; Larsbo et al., 2016). This complex relationship between SOC and PF may be attributed to soil properties (ratio of SOC to clay). It is not SOC, but complex SOC (complex of SOC and clay), that affects soil structure (Larsbo et al., 2016; Rahbeh, 2019), and research has shown that the SOC/clay composite threshold is 0.1 (Dexter et al., 2008). At low SOC and clay content (e.g., 0.5–2% SOC and 10–20% clay content), PF and SOC are positively correlated due to the formation of macropores when SOC/clay approaches or exceeds the composite threshold (Rahbeh, 2019). When the SOC content is higher, as the SOC increases, the number of macropores will instead decrease. High SOC leads to the formation of micron-sized pores, and these pores lead water to penetrate into the matrix, which is unconducive to the occurrence of PF (Larsbo et al., 2016; Rahbeh, 2019). Larsbo et al. (2016) also showed that as SOC increases, soil porosity will peak; beyond this peak value, if SOC continues to increase, soil porosity will instead decrease. The clay content of our selected observation stations is <8% (Fig. 6), and the complexing of SOC and clay will form large macropores, a scenario that is conducive to the occurrence of PF.

4.2.3. K_s

K_S also relates to water transport, because macroscopic pores have a great influence on K_S (Bouma, 1982). Mossadeghi-Björklund et al. (2016) found that K_S was proportional to the number of macropores and to the intensity of PF. We obtained similar results, that is, PF frequency was proportional to K_S ($r = 0.603$, $p < 0.01$). However, Etana et al. (2013) and Larsbo et al. (2016) found that low K_S was conducive to the occurrence of PF. Other studies performed under near-saturation conditions also found that PF was weaker in soil columns with greater porosity (larger K_S) (Larsbo et al., 2014; Katuwal et al., 2015; Paradelo et al., 2016). Larsbo et al. (2014) suggested that columns with smaller imaged porosity had lower near-saturated hydraulic conductivities, which presumably resulted in increased water pressures that were sufficient to generate flow in larger (but still percolating) pores. However, in cold, mountainous areas, it is difficult for soil moisture to approach saturation, and rainfall is poor, hence rainwater preferentially migrates through the large pores in the soil. Soil with large K_S has more macropores and better pore connectivity, which is conducive to the occurrence of PF (Guo and Lin, 2018).

4.3. Influence of initial soil moisture on PF

The classical assumption of the relationship between PF and initial soil moisture is that higher initial soil moisture is conducive to the occurrence of PF. This is due to the fact that higher initial soil moisture should lead to water pressure in the soil matrix approaching or exceeding the threshold water pressure, rendering PF more likely to occur under rainfall (Guo and Lin, 2018). However, the actual relationship between the two is rather complex. Many studies have found that the effect of initial soil water content on PF varies between different regions (Demand et al., 2019; Tang et al., 2020). We also found that, in different soil layers, the initial soil water content had different effects on PF (Fig. 9). For soil layers with higher sand content, the lower the soil moisture, the stronger the hydrophobicity, which is conducive to the occurrence of PF. However, in soils with lower sand content, wet soils are more prone to PF. Combined data indicate that for soils with macropore flow as the main type of PF, higher initial soil moisture is conducive to the occurrence of PF, while for soils that are susceptible to finger flow (such as sandy soil), drier precursor conditions are conducive to the occurrence of PF (Dekker and Ritsema, 1994; Guo and Lin, 2018; Demand et al., 2019).

5. Conclusion

We analyzed 2,288 infiltration events at eight soil moisture observation stations in the upper reaches of the HRB, in order to investigate the occurrence and controlling factors of PF in mountainous areas. PF frequency across all layers of these eight stations ranged from 0 to 22.28% (Due to differences in the methods for determining PF events, the frequency of PF occurrences should be considered as the lower bound). PF frequency decreased with increasing depth, while PF variability increased with increasing depth. The maximum PF frequency of our observation stations was lower than previous studies because we did not select forest stations and rainfall in our study area was low. Our results showed that spatial parameters (land cover and soil properties) and temporal factors (rainfall and initial soil water content) jointly controlled the occurrence of PF. The PF frequency in scrubland was the greatest, followed by grassland and meadow, and finally barren land. Low Bulk, low θr, high K_S, and high SOC were conducive to the occurrence of PF. Our results contrast with those in some previous studies. These contradictions can be explained by comparing the differences in soil properties and moisture conditions between different climatic regions and locations. We infer that there is a threshold for the influence of Bulk on PF; it has different effects on PF above and below this threshold. In cold, mountainous areas, where soil moisture is difficult to saturate and rainfall input is low, soil moisture observation stations with high K_S and SOC were conducive to the occurrence of PF. In addition, due to seasonal changes in rainfall, vegetation, and soil moisture, there were also seasonal differences in PF. The greater the cumulative rainfall, the greater the possibility of PF occurrence. However, due to the hydrophobicity of dry sand, we found that the effect of initial soil water content on PF changed in different locations. In areas with high sand content, dry soil was conducive to the occurrence of PF, while in areas with low sand content, the situation was reversed.

This study shows that the temporal and spatial factors have different effects on PF occurrence in different areas and under different initial conditions. Future research needs to be carried out in additional contrasting locations in order to better understand the impact of soil properties on PF.

CRediT authorship contribution statement

Weiming Kang: Conceptualization, Formal analysis, Methodology, Investigation, Writing – review & editing. **Jie Tian:** Conceptualization, Data curation, Project administration, Supervision, Writing – review & editing. **Yao Lai:** Software, Investigation, Validation, Writing – review &

editing. **Shaoyuan Xu:** Validation, Supervision, Data curation. **Chao Gao:** Supervision, Validation, Visualization. **Weijie Hong:** Software, Supervision, Validation. **Yongxu Zhou:** Software, Supervision, Data curation. **Lina Pei:** Investigation, Data curation, Visualization. **Chan-sheng He:** Project administration, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhydrol.2022.127528.

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